St Edward's College Malta

Year 10 Formal SBA 2025

LO13 Python Programming Task 1

Full Name

- This work is graded and contributes toward your SBA.
- The level of this task meets the criteria for SEC 09 Computing School-Based Assessment.
- The Assessment Rubric targets learing objectives at Level 1-2.
- The maximum mark awarded for this work is **27 marks**.

For teacher's use only:

Ms E. Camilleri

Criteria	1	2	3	Total
Obtained				
Allotted	15	6	6	27

For teacher's use only:

Assessment Rubric Task 1		
1. Python Programming Skills		
Variables for turns, guesses, and whether the player has guessed.	**3**	
Output messages for instructions, questions, and game play.	**3**	
Arithmetic calculations for counting turns and guesses.	**3**	
Conditional statements for handling player choices.	**3**	
Loop the program until the game is over.	**3**	
2. Programming Practices		
Neat code that is easy to read.	**3**	
Able to debug or troubleshoot errors without help.	**3**	
3. Bonus Marks		
No tokens were used and/or creative enhancements included and/or the program is complete and fully functional.	**3**	
Student used proper code nesting and has met criteria at level 3.	**3**	
Total	**27**	

General Comments:

Marking Guidelines

0 – Fails to meet the criteria or not attempted.

1 – Was able to complete the task by following instructions.

2 – Was able to complete the task with limited support.

3 – Was able to complete the task independently.

Task 1: Guess the Animal

Instructions:

Write a Python program that simulates a very simple Guess the Animal game!

The program should:

1. Display a title and instructions when the program starts.
2. Randomly choose one animal from a small list for example: `["cat", "dog", "elephant", "rabbit", "lion"]`.
3. A game has a maximum of 5 turns. In a turn, a player can ask a question to collect clues about the secret animal or make a guess. However, a player has up to 2 chances to guess.
4. End the game either when the player guesses correctly **or** runs out of turns.

Example Output:

```
********************
| GUESS THE ANIMAL |
********************
Welcome to Guess the Animal!
I have chosen a secret animal. You have 5 turns to figure out what it is!
In a turn, you can ask a question to get a clue or try to guess the
animal. Be warned: you only get up to 2 chances to guess!

Here are your options:
A. Does it live in water?
B. Is it big?
C. Does it have fur?
D. Does it roar?
E. I want to guess!

Turn 1 - Enter your choice: A
No.
Turn 2 - Enter your choice: C
Yes.
Turn 3 - Enter your choice: E
What’s your guess? lion
```

```
No, it is not a lion.
Turn 4 – Enter your choice: E
What’s your guess? cat
Yes. You win!
```

In the case that the player runs out of turns:

```
Turn 5 - Enter your choice: 5
What’s your guess? rabbit
No! The answer is cat. Game Over.
```

In the case that the player uses up the chances to guess in less than 5 turns:

```
Turn 1 - Enter your choice: A
No.
Turn 2 - Enter your choice: E
What’s your guess? Lion
No, it is not a lion.
Turn 3 - Enter your choice: E
What’s your guess? Rabbit
No! The answer is cat. Game Over.
```

Your final program must:

- Keep track of the number of turns and number of guesses in integer variables.
- Use at least two Boolean expressions: one to check whether the game is over and another one to check whether the player guessed.
- End the game if the player guesses correctly **or** uses up all the guesses **or** runs out of turns.
- Use conditional statements to give the appropriate clue based on the animal.

In the following page you will find a breakdown to help you build the program step-by-step and apply most of the Python skills covered in class.

Make sure to test your program after completing an exercise to check for syntax errors or bugs.

Task Breakdown:

1. Set Up the Game
    a. Display the title, welcome message, instructions and options as depicted in the example output provided.
    b. Create a list of animals: cat, dog, elephant, rabbit and lion.
    c. Use variables to store: the number of turns used, the number of guesses used, and whether the user guessed. You should initialize the number of turns used and guesses used to 0, and the "user guessed" variable to False.

2. Create the Game Loop
    a. Construct an infinite loop for the game to run.
    b. Write a condition statement that checks whether the game is over and breaks out of the infinite loop. The game is over if the player guesses correctly **or** uses up all the guesses **or** runs out of turns.
    c. During game play ask the user to enter their choice.

3. Handling Each Question
    a. When the player chooses a question (options A – D), display a "Yes" or "No" answer depending on the secret animal.
    b. When the player chooses to guess (option E), prompt the user to input an answer.
    c. Update the number of turns and guesses as appropriate.
    d. Use a Boolean expression to record whether the user has guessed the secret animal.

Appendix 1: Teacher Help Tokens

You have **TWO** help tokens for coding hurdles.

Having tokens helps balance your independence while providing support so that you can be successful. Tokens are not used to get answers, but to get guidance on how to move on, for example, pointing you to relevant material or helping you get organized. Should you manage to complete the task to a high standard without any use of tokens, you get awarded bonus marks!

To use a token, raise your hand, and when the teacher approaches you, redeem a single token.

Help Token	**Help Token**

Appendix 2: Python Code Constructs

- Comment: `# This is a comment`

- Output:

Example 1

```
print("This is a message")
```

Example 2

```
print(message)
```

Example 3

```
print("Message:", message)
```

- Input: `user_name = input("Enter name:")`

- Variables and arithmetic:

Assignment (=)	`a = 0`
Addition (+)	`a = a + 1`
Subtraction (-)	`a = a - 1`

- Conditional operators:

Equal to	`a == b`
Not equal to	`a != b`

- Logical operators: `and`, `or`, `not`

- Conditional statement:

Example 1

```
x = 2
if x == 2:
    print("x is two!")
else:
    print("x is not two.")
```

Example 2

```
x = 2
if x == 1:
    print("x is one")
elif x == 2:
    print("x is two")
else:
    print("x is not valid!")
```

Example 3

```
name = "John"
age = 23
if name == "John" and age == 23:
    print("Ok, verified.")
else:
    print("Unverified.")
```

- Infinite loop with break:

```
count = 1
while True:
    print(count)
    count = count + 1
    if count >= 5:
       break
```

- List:

```
my_list = [12, “Jack”, 26]
```

- Using the Random module to get a random choice from a list:

```
import random # at the very top of your program
secret = random.choice(my_list)
```

-- End of SBA --